Cloud Computing: Practical Applications in the Modern World

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UNIT – 1.1.1

Overview of cloud computing: Definition, history, and evolution. Cloud service models: IaaS, PaaS, SaaS. Cloud deployment models: Public, private, hybrid, and community clouds. Key features: Scalability, elasticity, on-demand provisioning, and multi-tenancy. Virtualization: Hypervisors, containerization (Docker), and orchestration (Kubernetes).

On-Premise Infrastructure

On-premise infrastructure refers to IT resources physically located within an organization, providing greater control, security, and customization compared to cloud solutions.

On-premise infrastructure, often referred to as "on-prem," consists of computing resources such as servers, storage, and networking equipment that are physically housed within an organization's facilities.

Cloud computing

Distributed computing on internet or delivery of computing service over the internet.

Eg: Yahoo!, GMail, Hotmail

Instead of running an e-mail program on your computer, you log in to a Web e-mail account remotely. The software and storage for your account doesn't exist on your computer -- it's on the service's computer cloud.

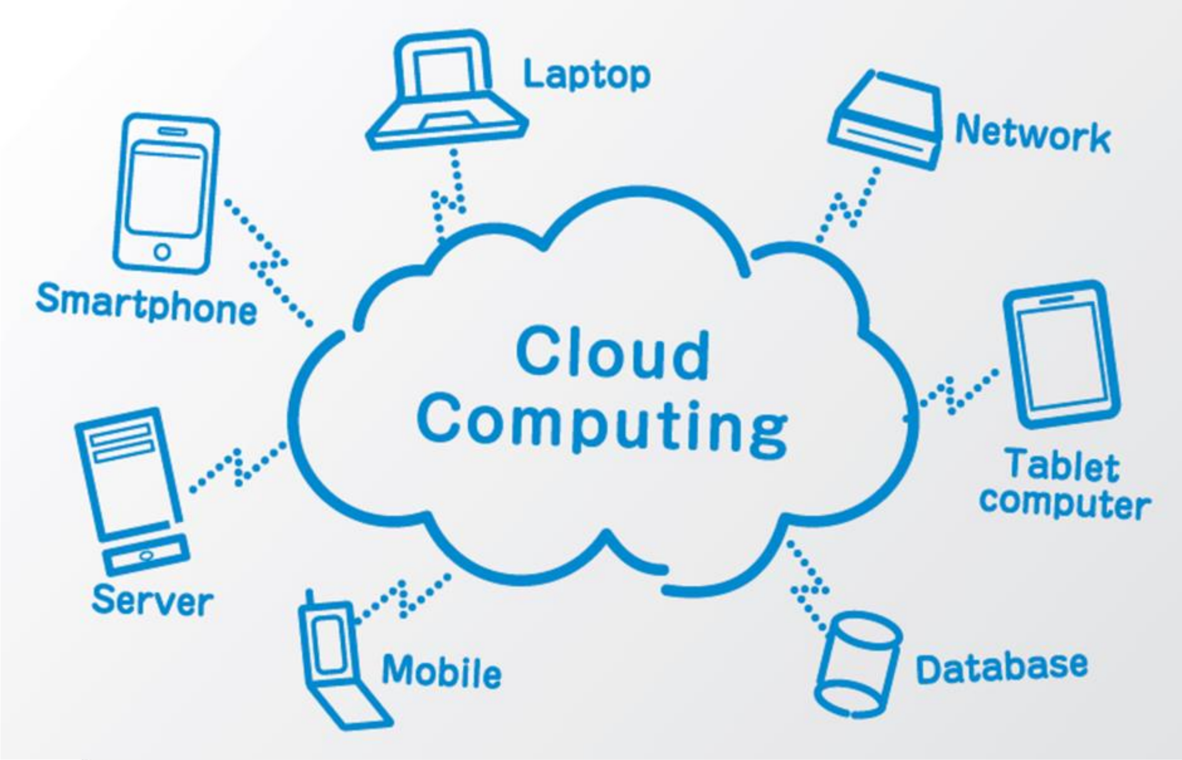
Cloud computing

Cloud Computing refers to the delivery of computing services—including servers, storage, databases, networking, software, and more—over the Internet ("The Cloud").

Key Attributes:

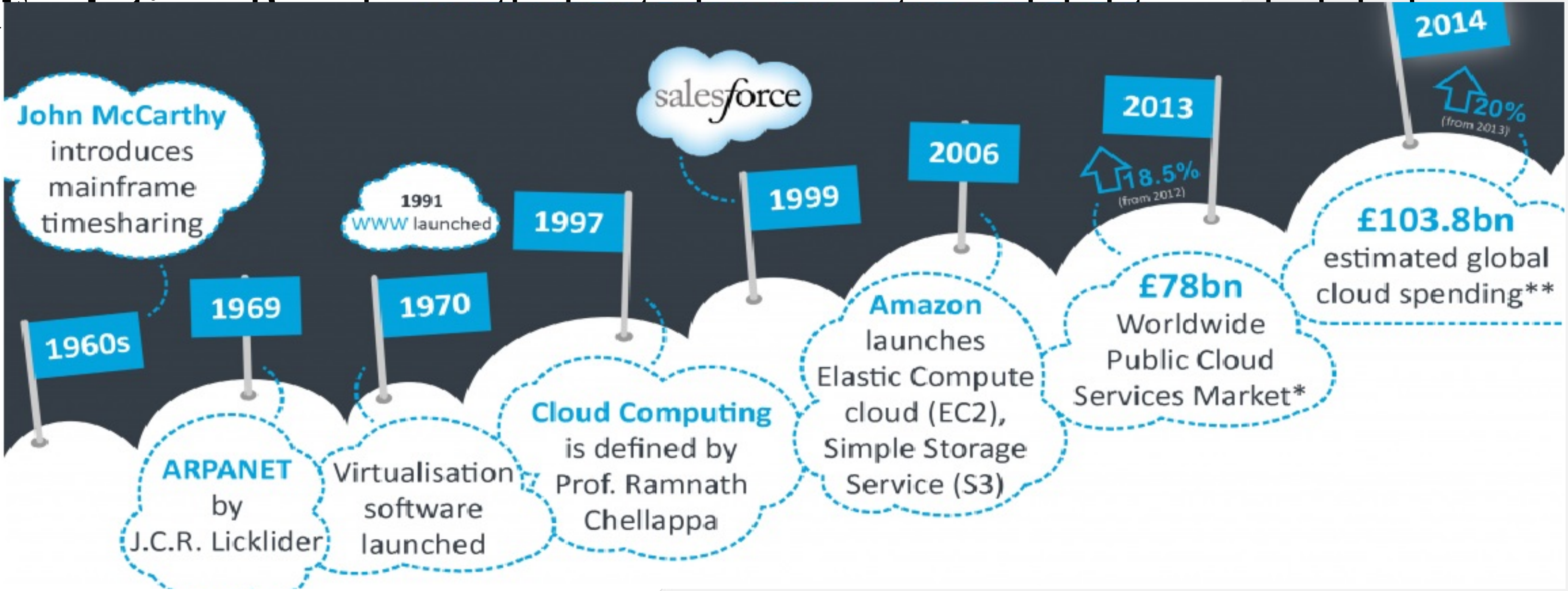
- Cost-efficient
- Scalable
- Accessible from anywhere
- Pay-as-you-go

[Cloud Computing | InnoAge-Technologies PVT. LTD.]



Cloud history, and evolution

- Concept evolved in 1950 (IBM) called RJE (Remote Job Entry Process).
- In 2006 Amazon provided First public cloud AWS (Amazon Web Service).



Top 10 **Cloud Providers**

aws

Azure

Google Cloud

Alibaba Cloud

salesforce

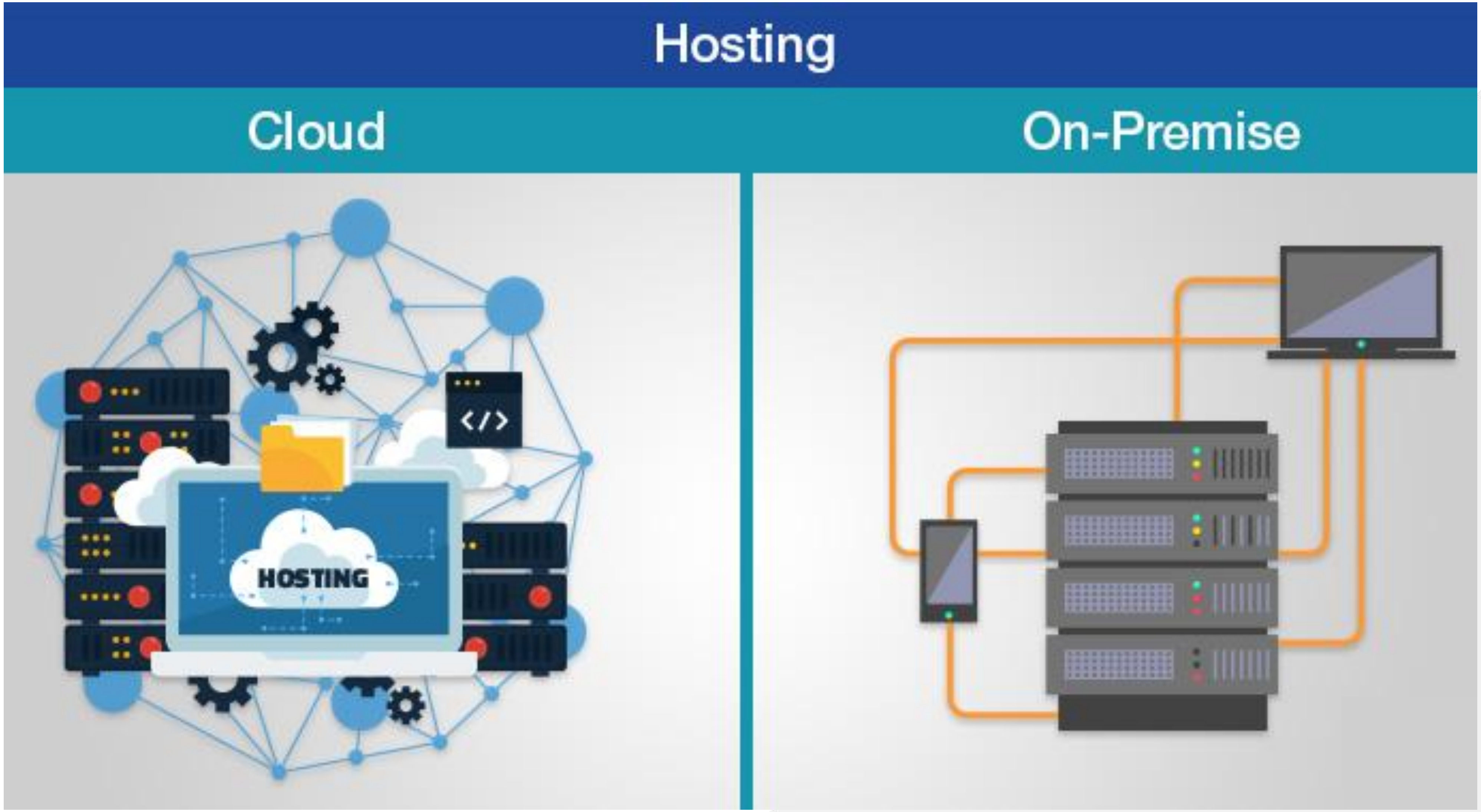
DELL

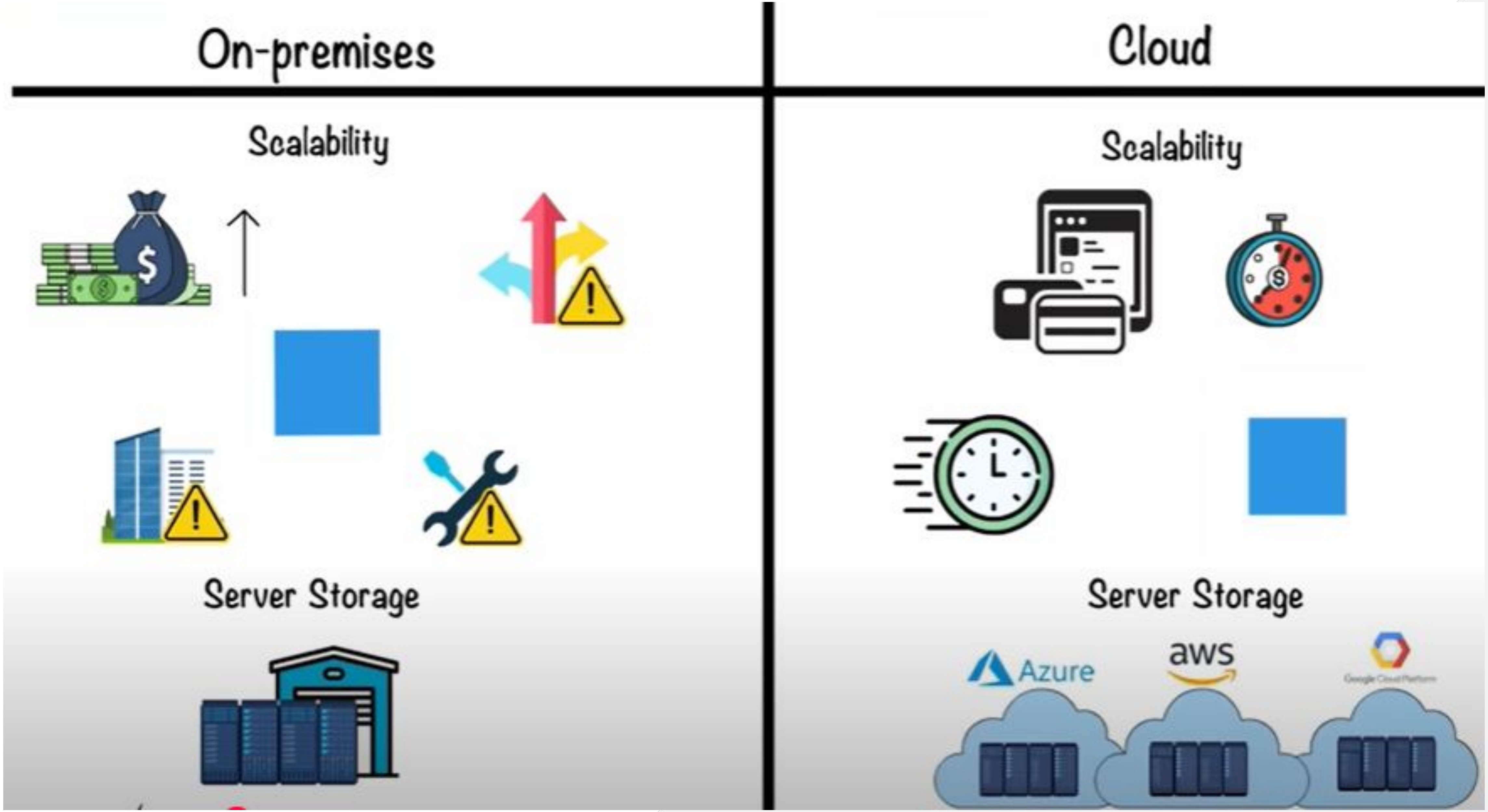
IBM

DigitalOcean

Adobe Creative

Dropbox


Hosting
Cloud
On-Premise
HOSTING


On-premises
Cloud
Scalability
Scalability
Server Storage
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Azure
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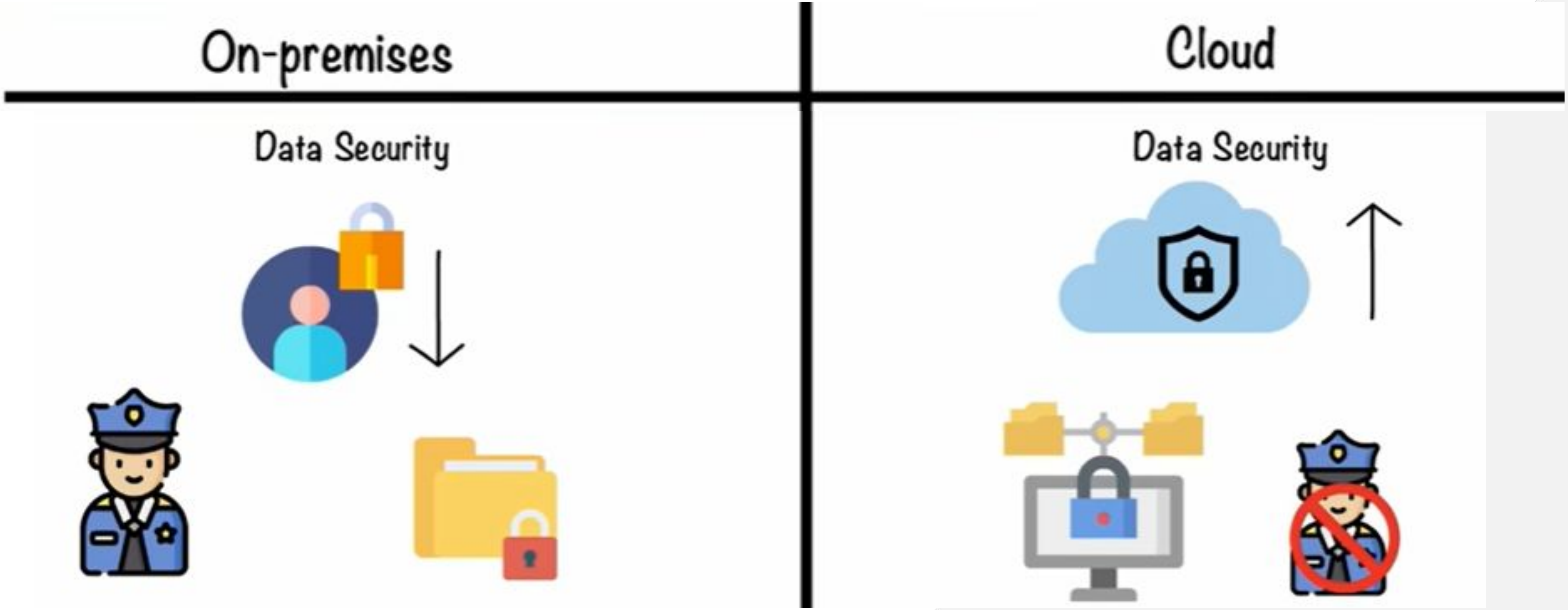
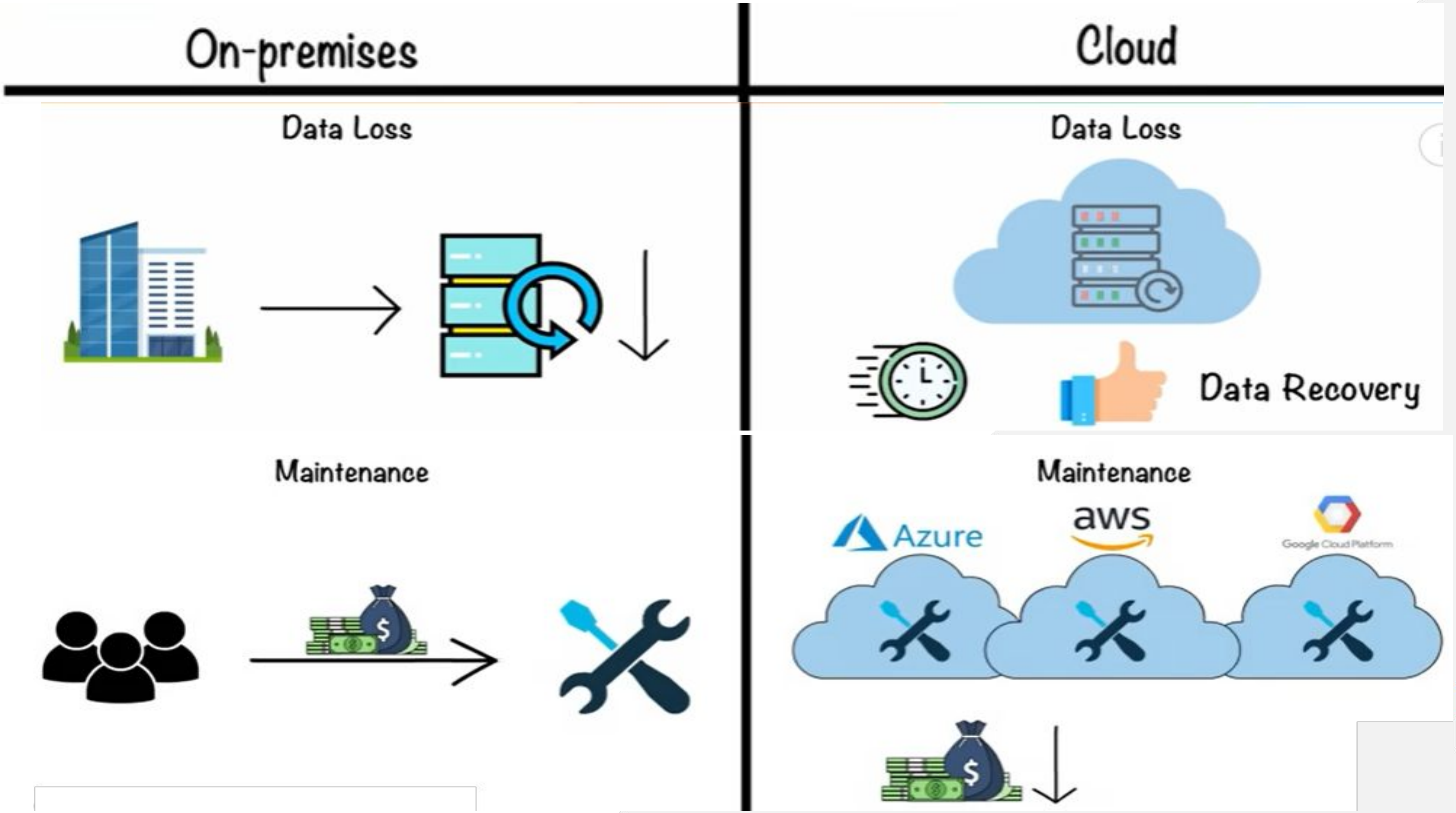
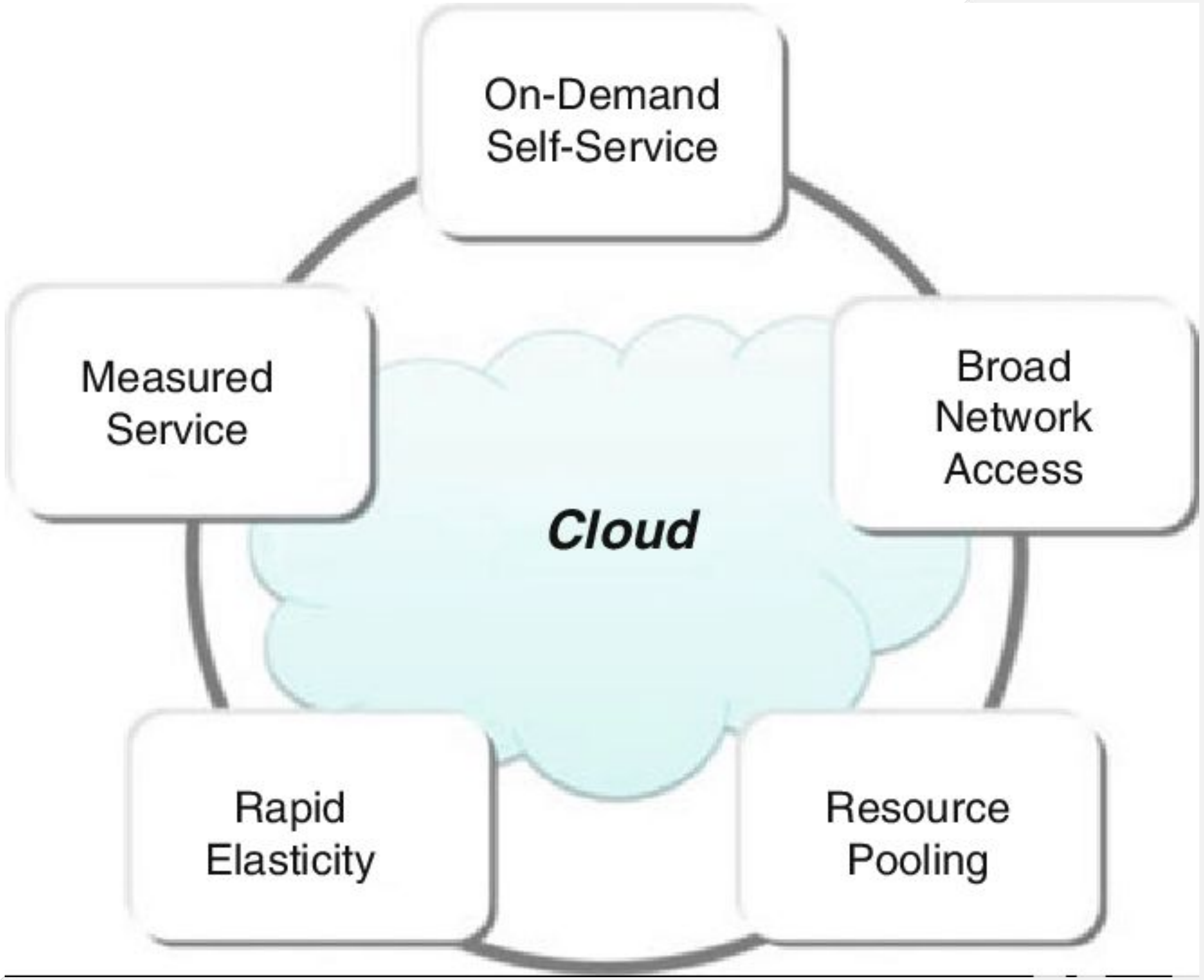

On-premises
Cloud
Data Security
Data Security


On-premises
Cloud
Data Loss
Data Loss
Data Recovery
Maintenance
Maintenance
Azure
aws
Google Cloud Platform

Key Features of Cloud Computing

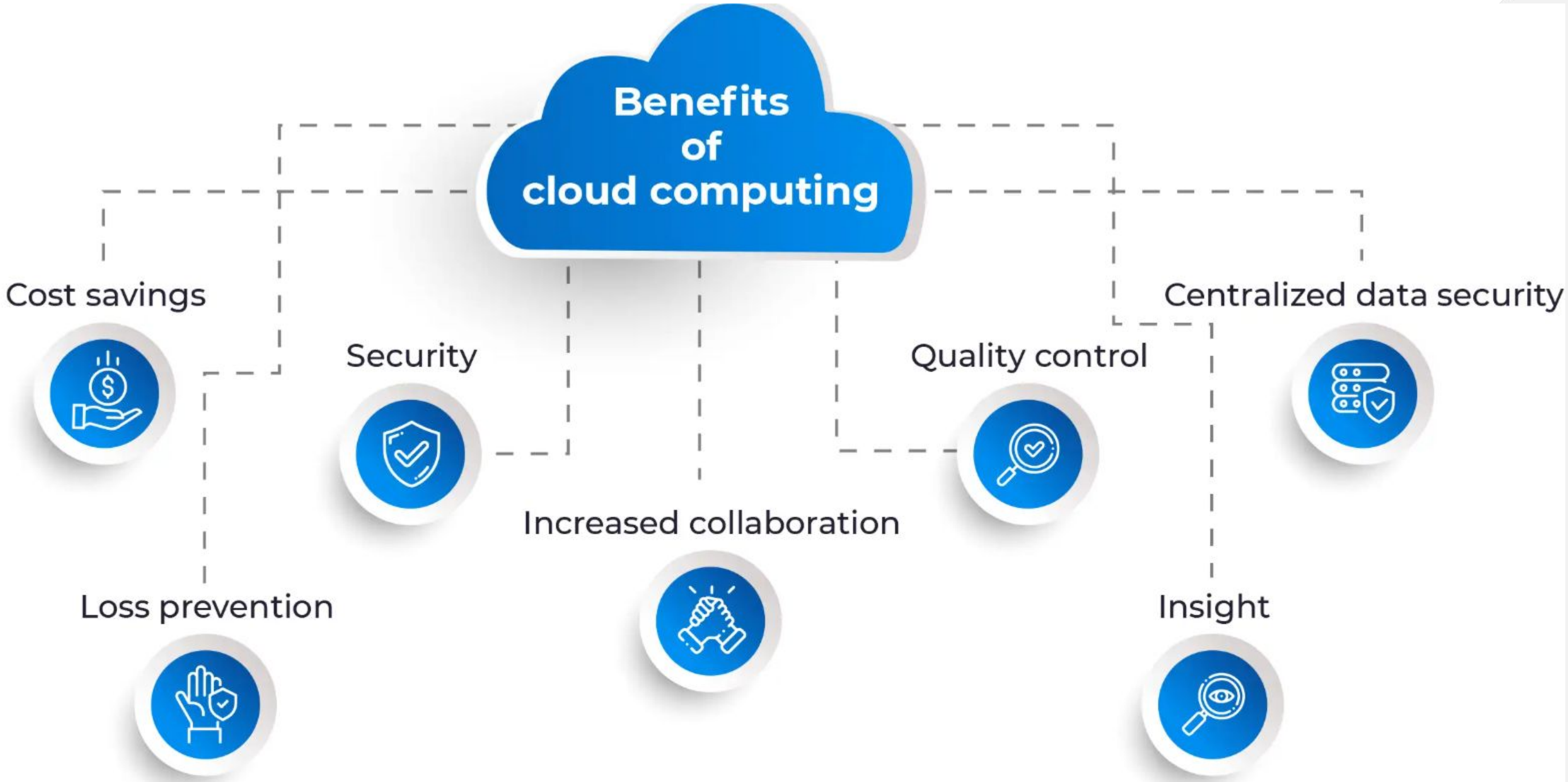
- **Scalability:** Ability to handle growing amounts of work or traffic.
- **Elasticity:** Automatically scale resources up or down based on demand.
- **On-Demand Provisioning:** Immediate resource allocation without manual intervention.
- **Multi-Tenancy:** Multiple users share the same infrastructure securely.

[Characteristics of Cloud Computing | GeeksforGeeks]

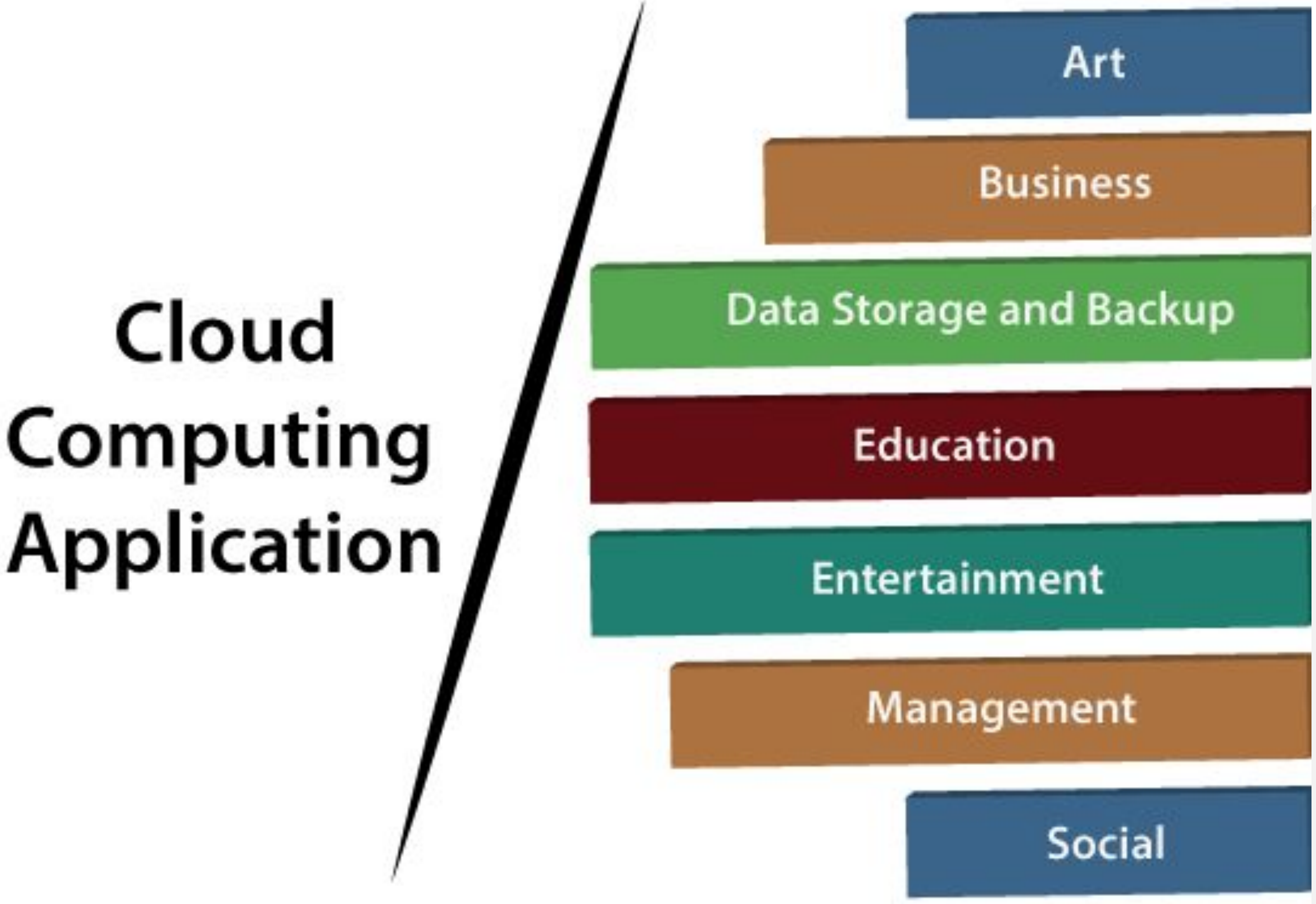


Key Features of Cloud Computing

Benefits of Cloud Computing



Applications of Cloud Computing

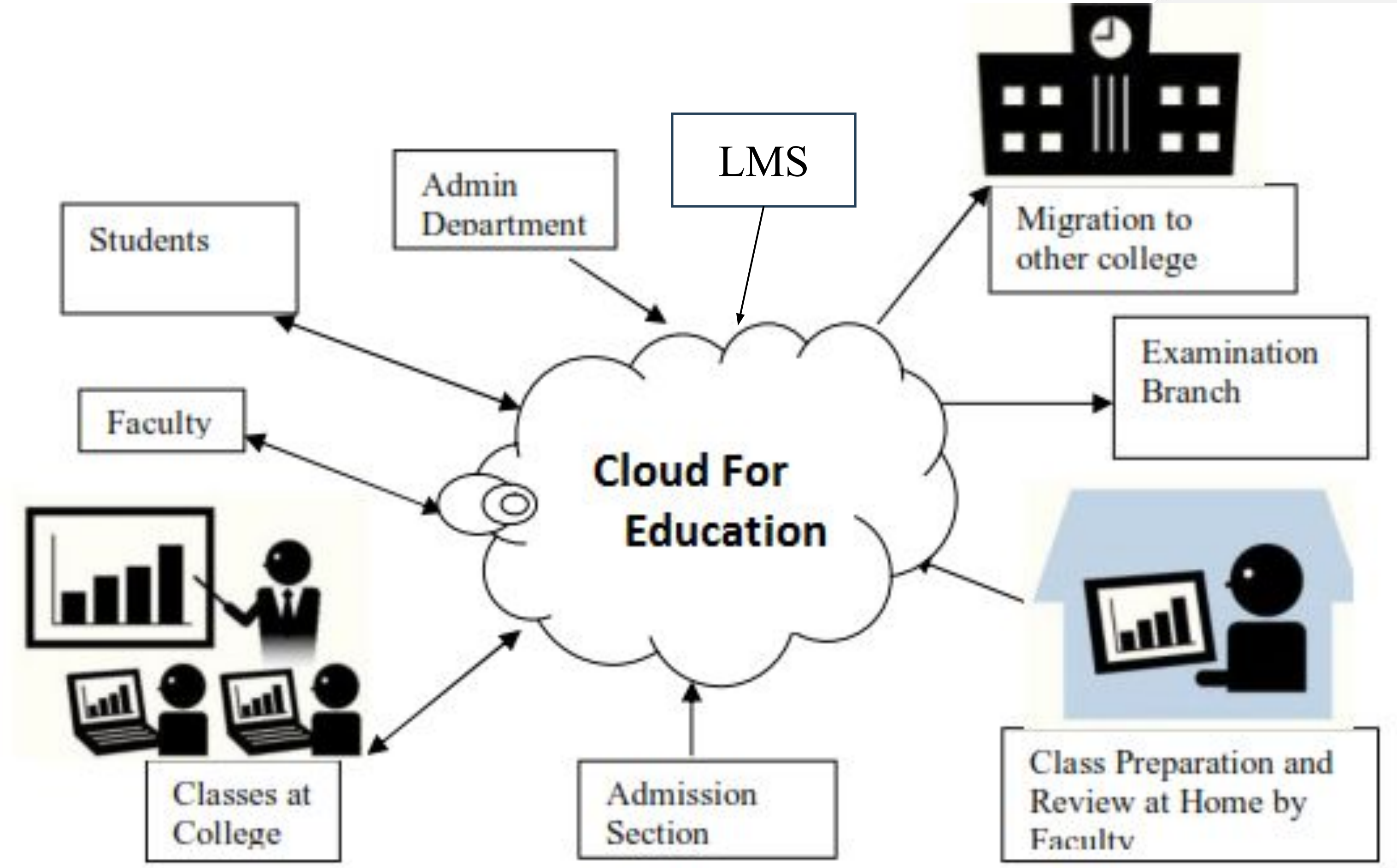


Cloud in Healthcare

- Patient data management. For example - *Health Insurance Portability and Accountability Act* (HIPAA) - compliant storage)
- Telemedicine platforms hosted on cloud
- AI-driven diagnostics using cloud compute
- Example: Philips HealthSuite on AWS

Cloud in Education

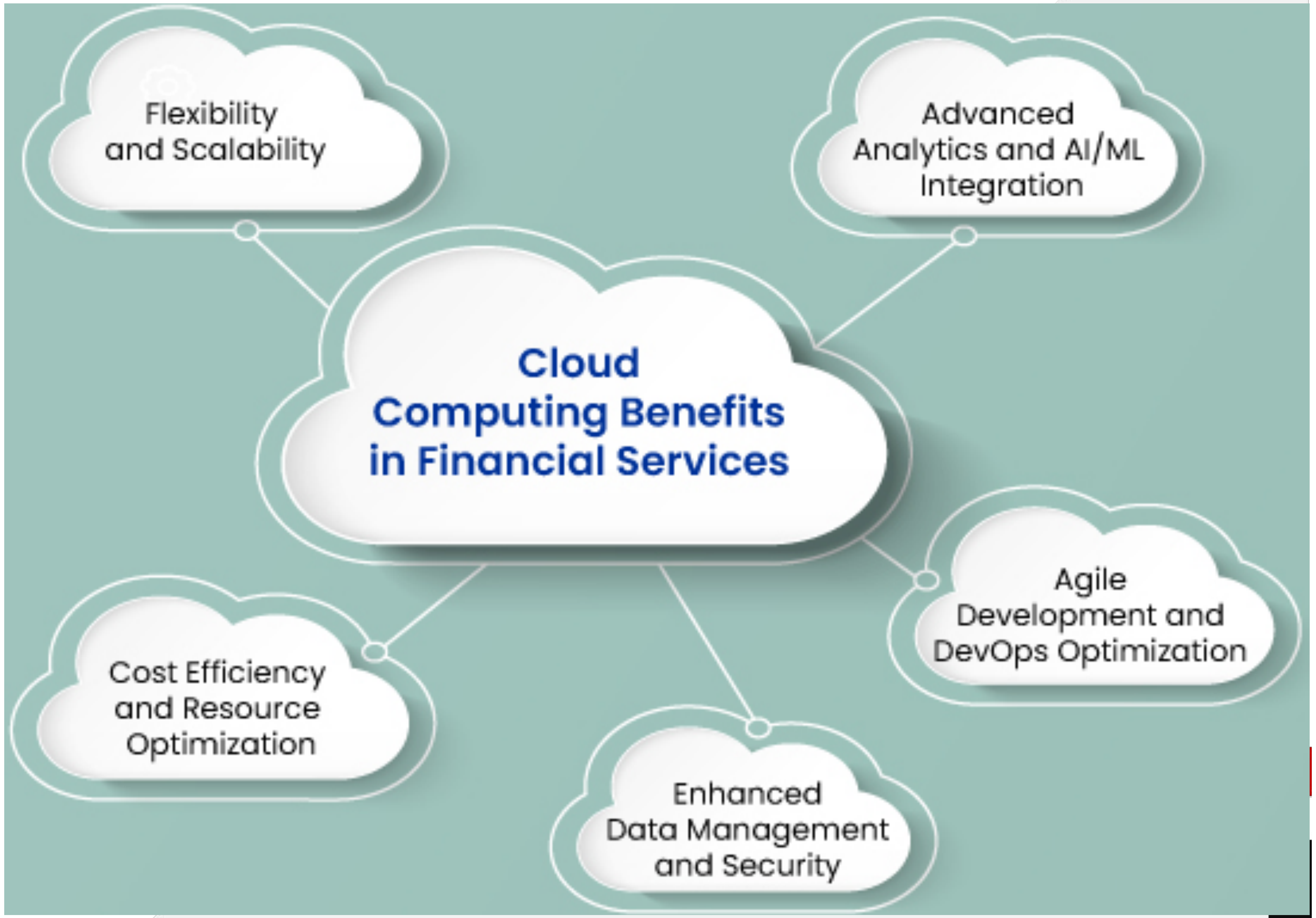
- Online learning platforms (e.g., Google Classroom, Coursera, AWS Educate)
- Virtual labs and sandbox environments
- Scalable LMS and ERP systems for institutions
- Real-time collaboration tools (Zoom, Microsoft Teams, etc.)



Services attached to Education Cloud

Cloud in Finance

- Secure banking apps using cloud infrastructure
- Fraud detection using cloud-based ML models
- High availability and scalability for transaction platforms
- Compliance (e.g., *Payment Card Industry Data Security Standard* (PCI DSS) in cloud setups)

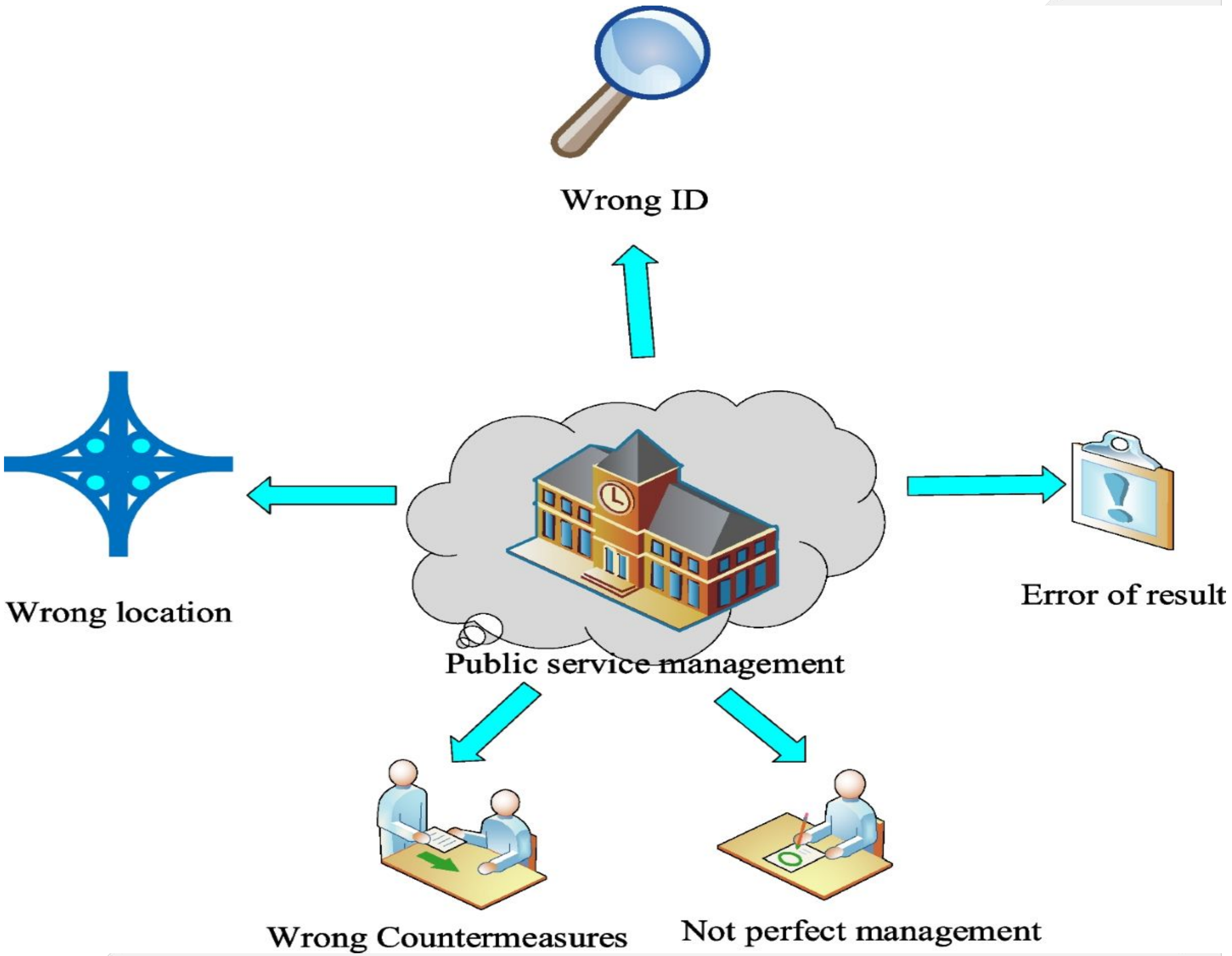


Cloud in E-Commerce

- Hosting scalable online stores (e.g., Shopify, Magento on AWS/Azure)
- Real-time inventory and logistics
- Personalization using cloud analytics
- Example: Netflix using AWS for content delivery

Cloud in Government and Public Services

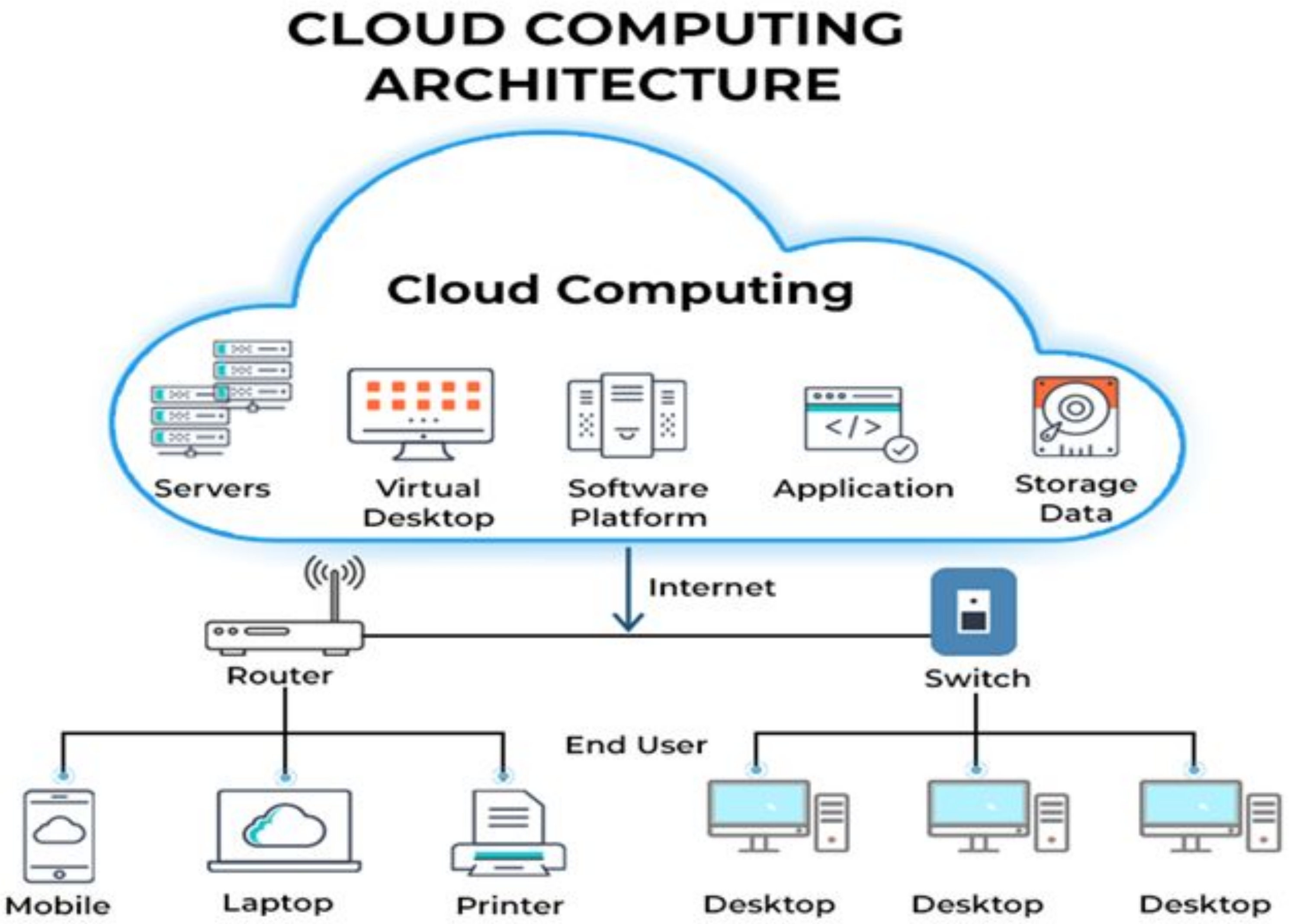
- e-Governance portals (e.g., DigiLocker)
- Smart cities infrastructure
- Disaster response systems
- Open data platforms



Cloud & Emerging Technologies

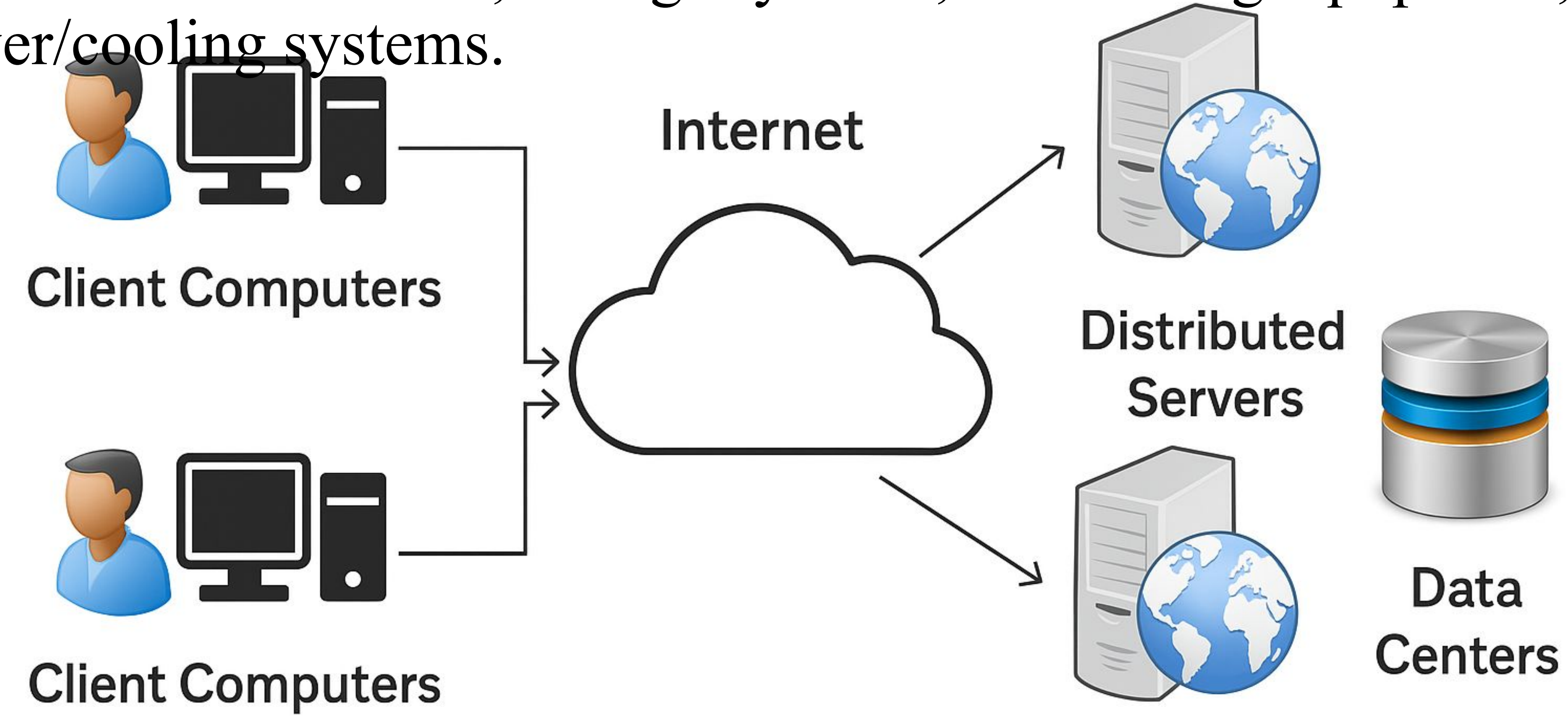
- AI & ML model training and inference
- IoT platforms and edge-cloud integrations
- Big Data processing with cloud services (e.g., AWS EMR - Amazon Elastic MapReduce, Azure Synapse)
- Blockchain-as-a-Service

How does cloud computing work?



Cloud computing Components

- **Client Computers** – End users interactive devices – Mobile, Laptops etc.
- **Distributed Servers** – geographically distributed servers – acts as working next to each other.
- **Data Centers -** physical facilities that house the critical infrastructure—servers, storage systems, networking equipment, and power/cooling systems.



How does cloud computing work?

- Cloud computing works by enabling client devices' data and cloud applications over the internet from remote physical servers, databases, and computers, network, and cloud software applications, with the back end, which consists of databases, servers, and computers.
- The backend functions as a repository, storing data that is accessed by the front end.
- Communications between the front and back ends are managed by a central server. The central server relies on protocols to facilitate the exchange of data.
- The central server uses both software and middleware to manage the connectivity between different client devices and cloud servers. Typically, there is a dedicated server for each application or workload.

Why cloud service is popular?

- Reduce the complexity of networks.
- Do not have to buy software licenses.
- Customization.
- Scalability, reliability, and efficiency.
- Data at cloud are not easily lost.
- Cloud providers that have specialized in a particular area (such as e-mail) can bring advanced services that a single company might not be able to afford or develop.

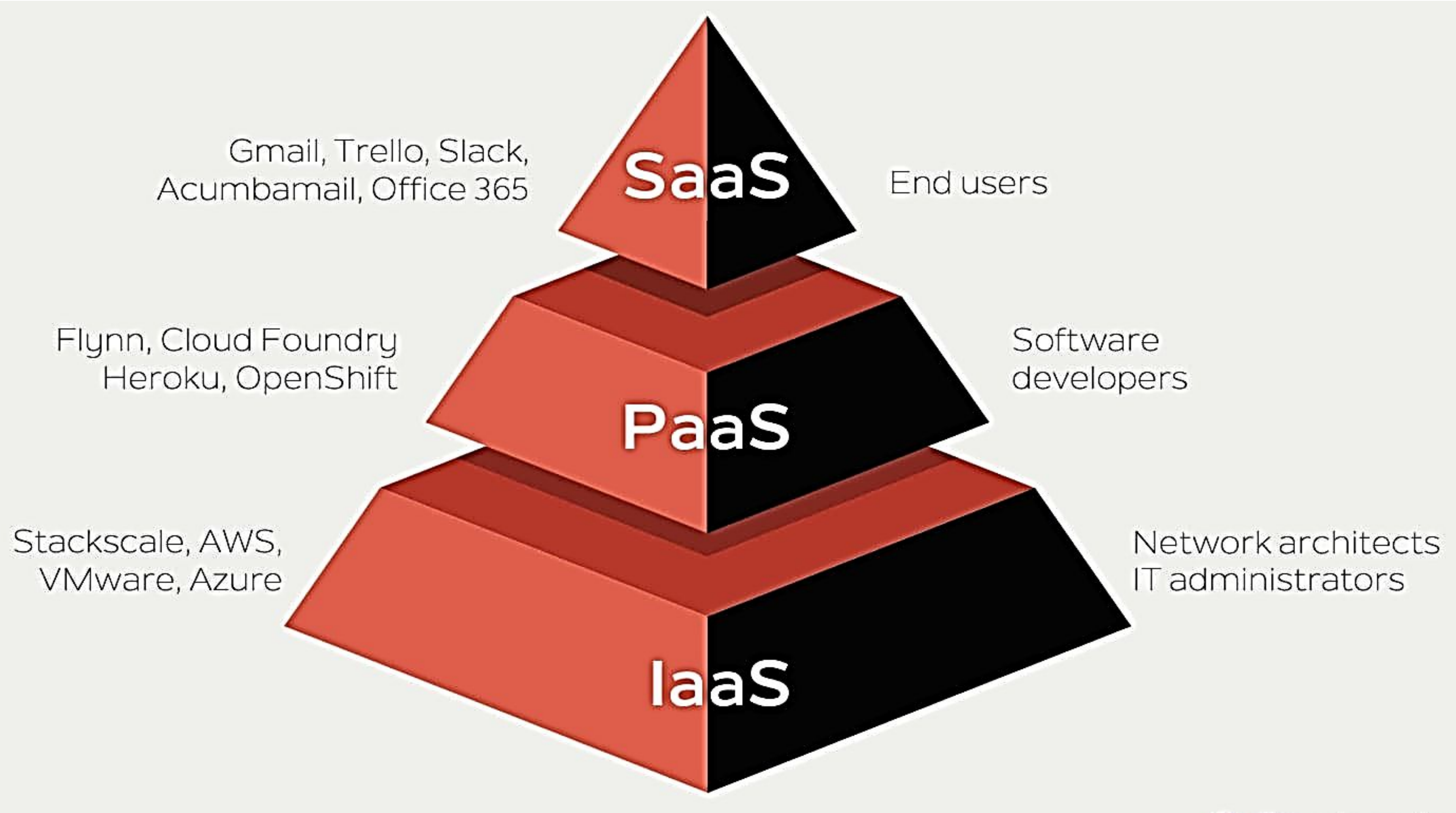
Cloud Service Models

IaaS (Infrastructure as a Service):
Provides virtualized computing resources over the internet.
Examples: AWS EC2, Google Compute Engine.

PaaS (Platform as a Service):
Provides a platform to develop, run, and manage apps.
Examples: Google App Engine, Heroku.

SaaS (Software as a Service):
Delivers software applications over the internet on demand.
Examples: Gmail, Microsoft 365.

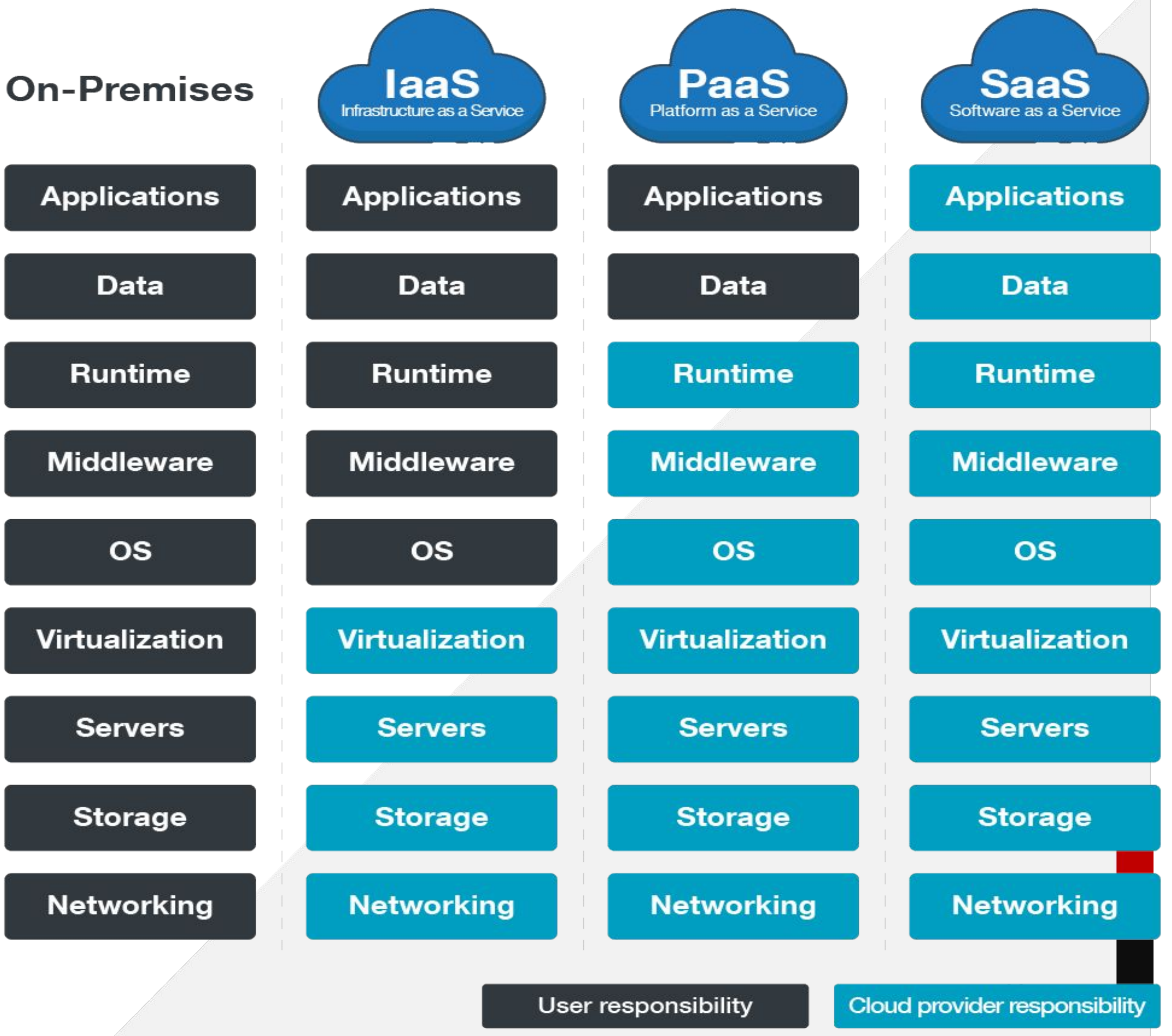
Cloud Computing Service Models



[Cloud Service Models - javatpoint]

Cloud Computing Service Models

- **Applications** – Custom or third-party software.
- **Data** – Databases, files, and information storage.
- **Runtime** – Execution environment for applications.
- **Middleware** – Software that connects applications and services.
- **Operating System (OS)** – System software managing hardware and software resources.
- **Virtualization** – Abstracting hardware to create virtual machines.
- **Servers** – Physical machines hosting applications and data.
- **Storage** – Data storage infrastructure.
- **Networking** – Connectivity between systems



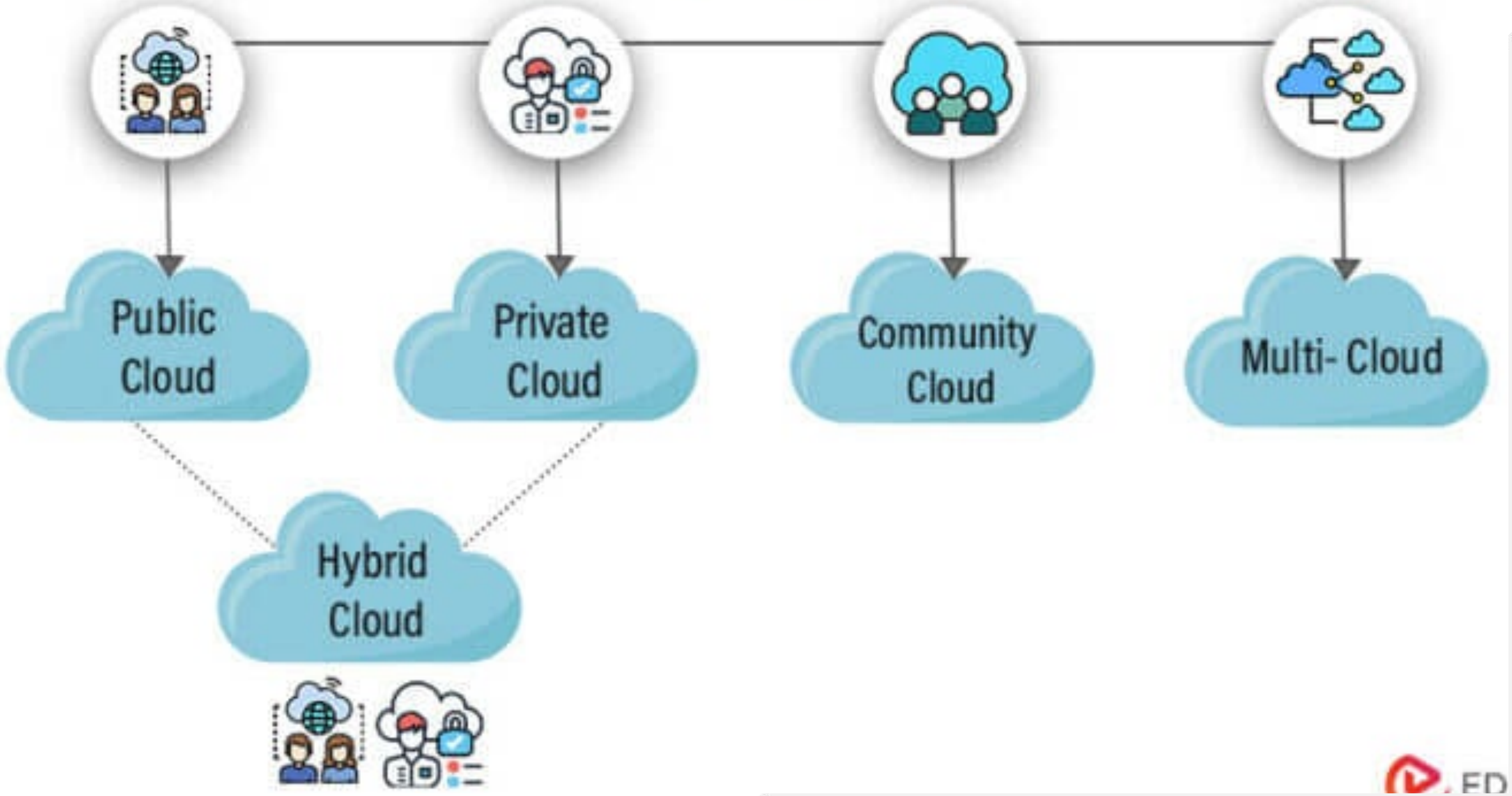
Difference between IaaS, PaaS, and SaaS

IaaS	PaaS	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform +Software.

Cloud Deployment Models

Cloud deployment models are used to determine where infrastructure resides, who controls it, and how it is accessed, allowing organizations to balance cost, security, and scalability. They enable flexible, on-demand access to resources—such as public, private, or hybrid clouds—without significant upfront hardware investment.

Cloud Deployment Models



Cloud Deployment Models

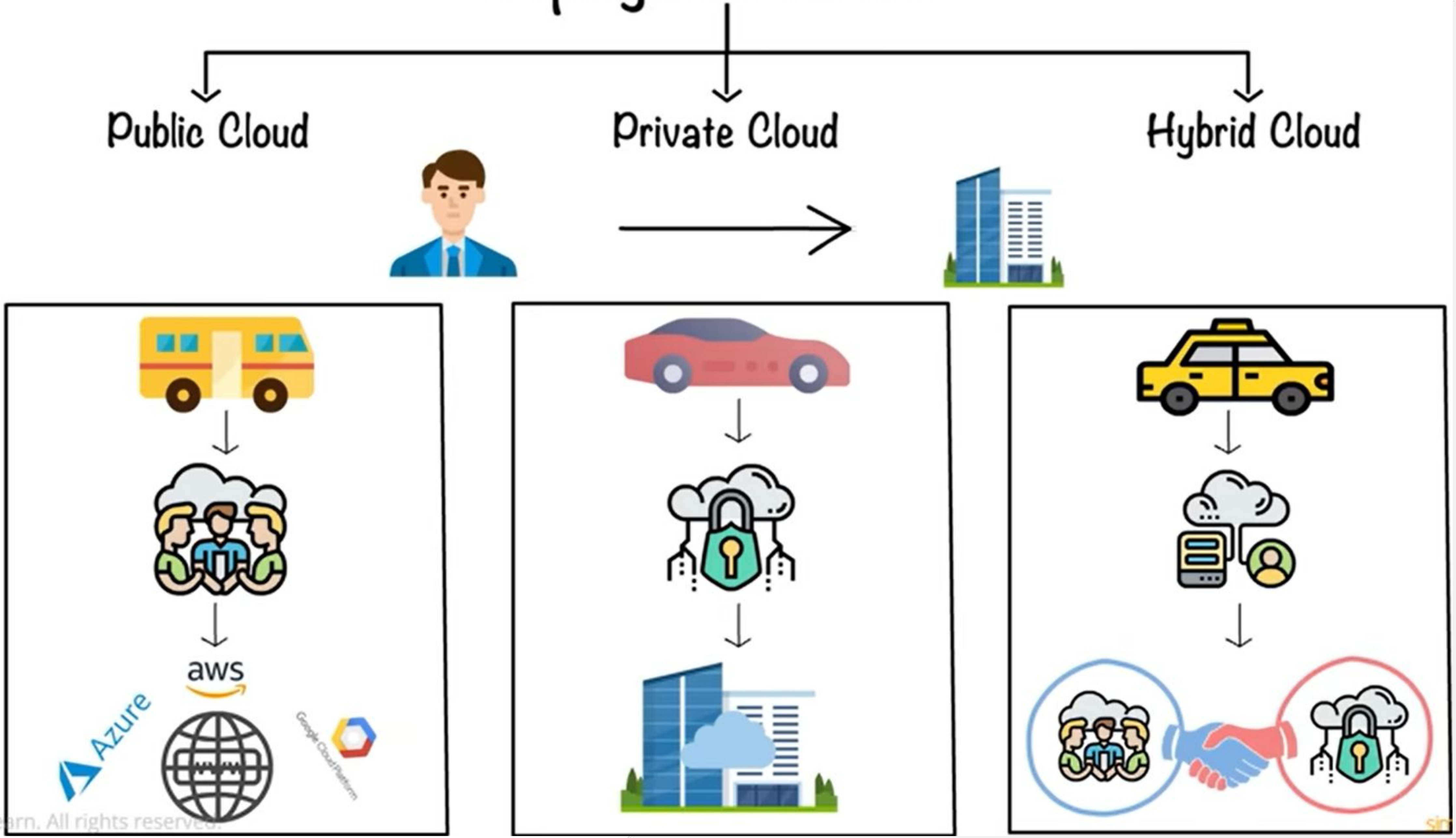
Public Cloud:
Services offered over the public internet and available to anyone authorized.
Examples: AWS, Azure.

Private Cloud:
Exclusive cloud environment for a single organization.
More control, security, and customization.

Hybrid Cloud:
Combines public and private clouds, allowing data/apps to move between them.

Community Cloud:
Shared infrastructure for organizations with similar concerns (e.g., security, compliance, media).

Deployment Model



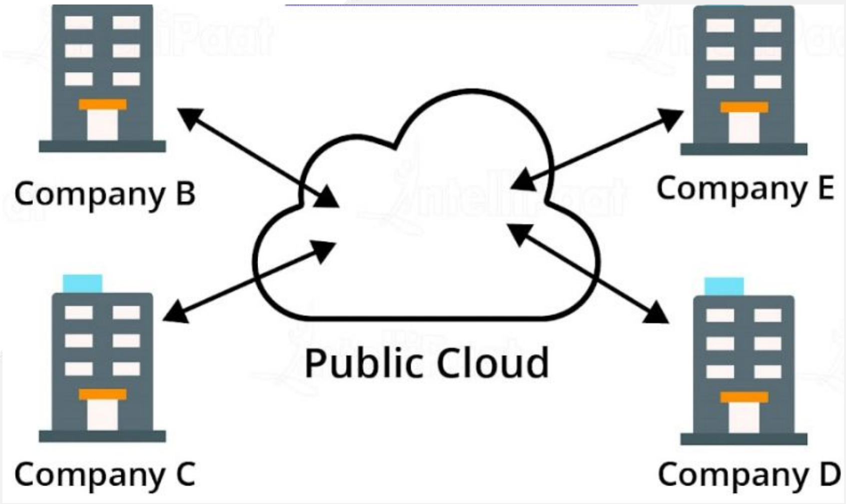
Public Cloud

✔Anybody can access systems and services

✔Less secure as it is open to everyone

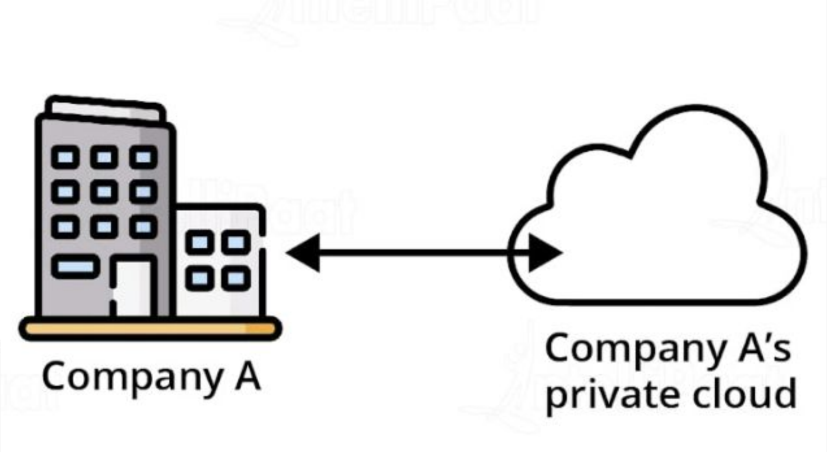
✔Maintenance work is done by the service provider

✔Example – Gmail, Google drive, Netflix, Spotify, Dropbox



Private Cloud

- ✔one-on-one environment for a single user
- ✔Implemented in a cloud-based secure environment protected by **firewalls**
- ✔Suitable for storing corporate information to which only authorized staff have access
- ✔Example – VMware, CUIMS, Banking Systems, Healthcare Providers, Government Agencies



Hybrid Cloud

✔Combination of public and private cloud

Hybrid = Public + Private

✔Company can balance the load by locating mission-critical workloads on private cloud and deploying less sensitive ones on public cloud

✔Example – Uber, GE Aviation, Retail Companies, Manufacturing, Educational Institutions

Issues related to Cloud Computing

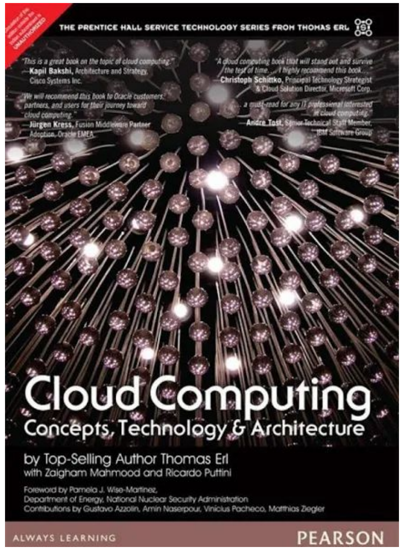
- **Security** – chances of data breach
- **Downtime** – Regular internet required
- **Vendor lock-in –** can’t easily switch from one cloud service provider to another
- **Latency -** Cloud relies on internet connectivity, which can introduce latency (delay)
- **Data loss** - Users might accidentally delete data would result in data becoming inaccessible

Future of Cloud Computing

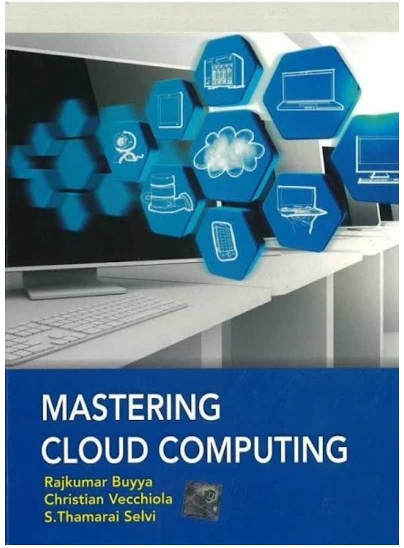
- Rise of serverless and Function-as-a-Service (FaaS)
- AI-optimized infrastructure
- Sustainability and green cloud computing
- Continued rise of hybrid and multi-cloud strategies

For more insight

Textbooks



Cloud Computing: Concepts, Technology & Architecture; Pearson Publication



Mastering Cloud Computing; Tata McGraw-Hill Education